



EXPERIENCES WITH GRINDING AND MILLING IN NORTH AMERICA – A SUPPLIER'S VIEW

B O S T O N



Refresher – Grinding and Milling



Grinding and Milling - Basics

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Rail Grinding

- Abrasive (dry) process with rotating grinding stones
- Face grinding technology: Rotational axis perpendicular to rail surface

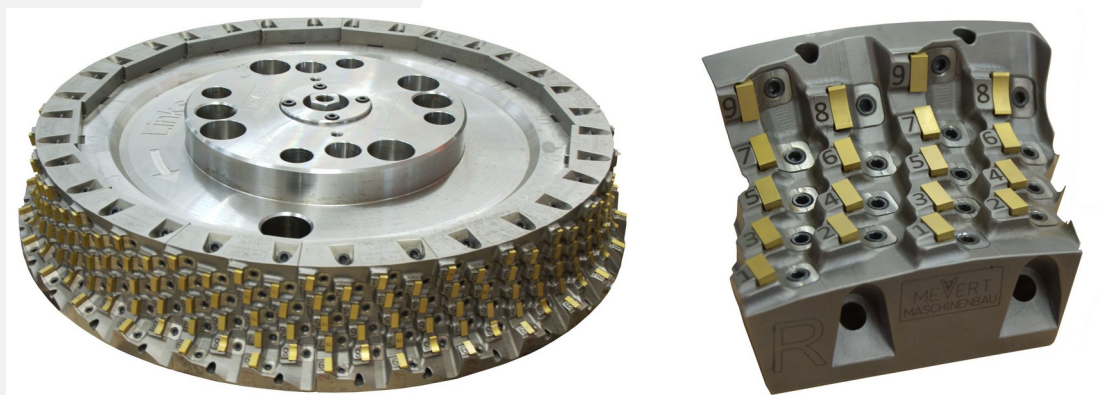


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B O S T O

Rail Milling

- Non – abrasive, dry cutting process with rotating cutter heads (carbides)
- Peripheral milling: Rotational axis parallel to rail surface



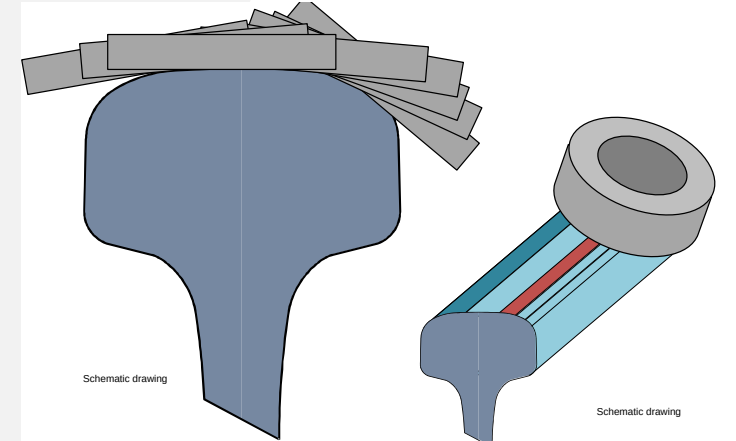


Grinding and Milling: Transversal Profile

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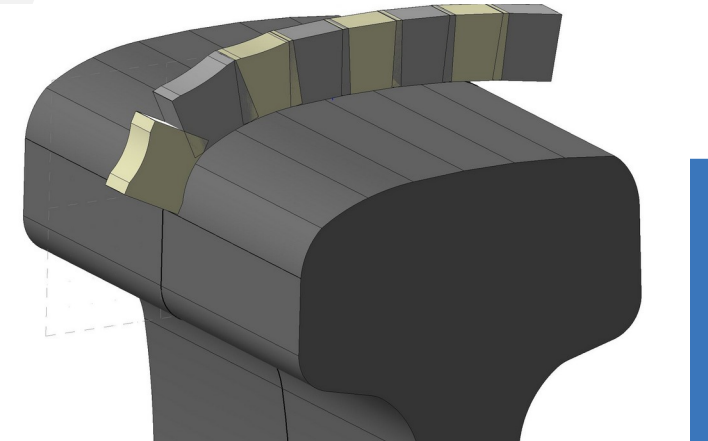
Rail Grinding:

- Flexible profile adjustment through angular stone alignment



Rail Milling:

- Profile defined by negative shape of cutter head (milling tool) with carbide inserts





Grinding and Milling - Results

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Resultant product looks similar, but milling $\neq$ grinding





The Supplier's Journey

B O S T O



The Starry-Eyed Supplier

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- North America: Rail milling white spot
- Rail Milling successful in Europe, Asia and to some extend Australia
- Europe: 165,000 mi
- USA + CAN: 177,000 mi
- Extrapolation for milling machine potential





The Best Invention Since Sliced Bread

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- Common idiom used to praise an invention or development
- A few words about “Sliced Bread”
- Bread slicing machine invented 1912 by Otto Frederick Rohwedder in Davenport/Iowa
- Took 16 years to get traction (1928)





The Best Invention Since Sliced Bread

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- Corrective / regenerative maintenance
 - Remove a lot of metal in one pass
- High quality rail surface
- Spark and dust free
- By-product collection
- Implemented in Europe and Asia
- Only preventive grinding
- Milling removes too much metal
- Milling is sooo slow
- We have 4 different target profiles
- Happy with current quality
- What's the value of dust free



The Sobered Supplier

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- Rail is Rail, but there are significant differences between continents
 - Technical, legislative, economical and cultural
- Grinding is engrained as maintenance technology
 - The only technology for 50+ years
 - Some technical barriers
 - Monopolistic (grinding) situation



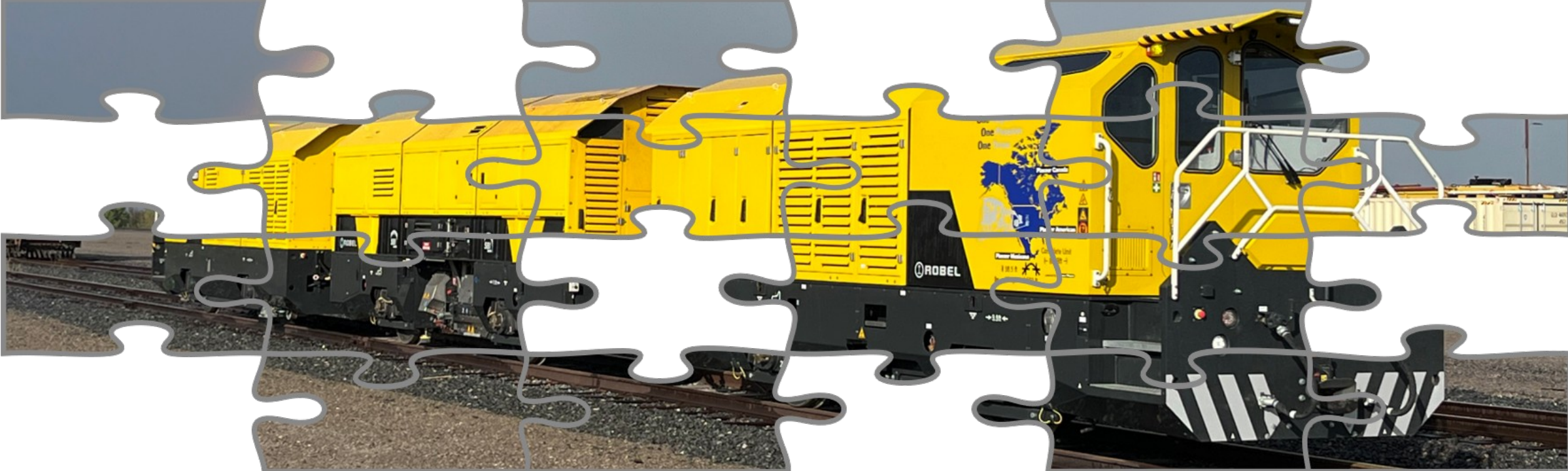


The Educating Supplier

Challenge: Would you trust your supplier educating you about his products?

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The Solution to the Challenges



Milling Removes too Much Metal

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- Milling can remove 0.3mm or more per pass
- Max. cutting depth (per pass):
 - Machine dependent 1.0 – 6 mm
 - Transit (small machine): 1-1.5mm
 - Controllable in 0.1mm steps
- Theoretically min. cutting depth of 0.1mm
 - Challenging due to elasticities of track and machine

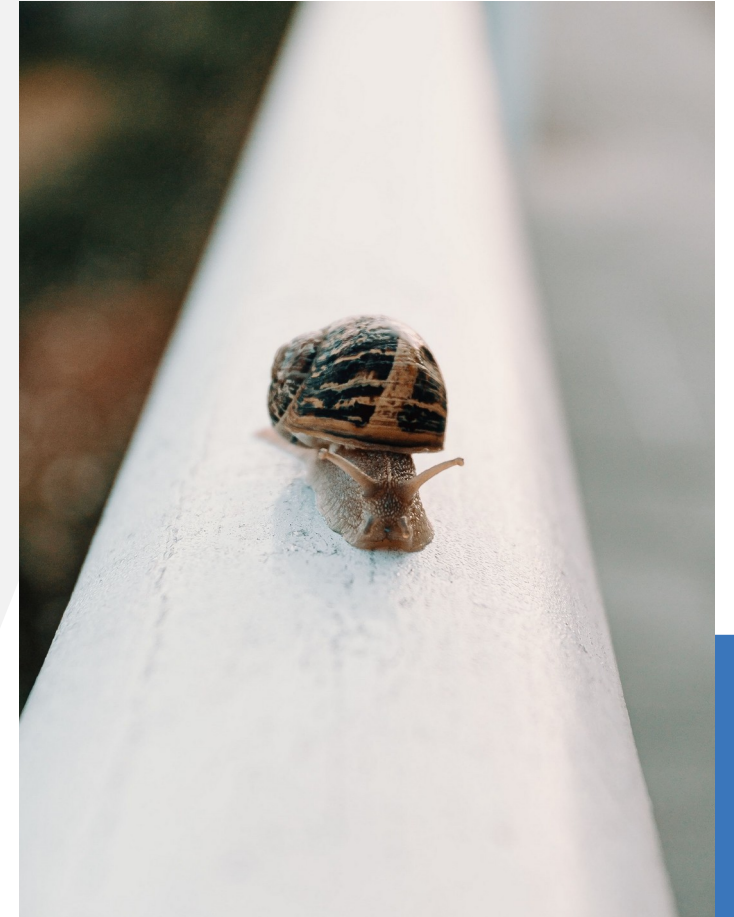




Milling is Sooooo Slow

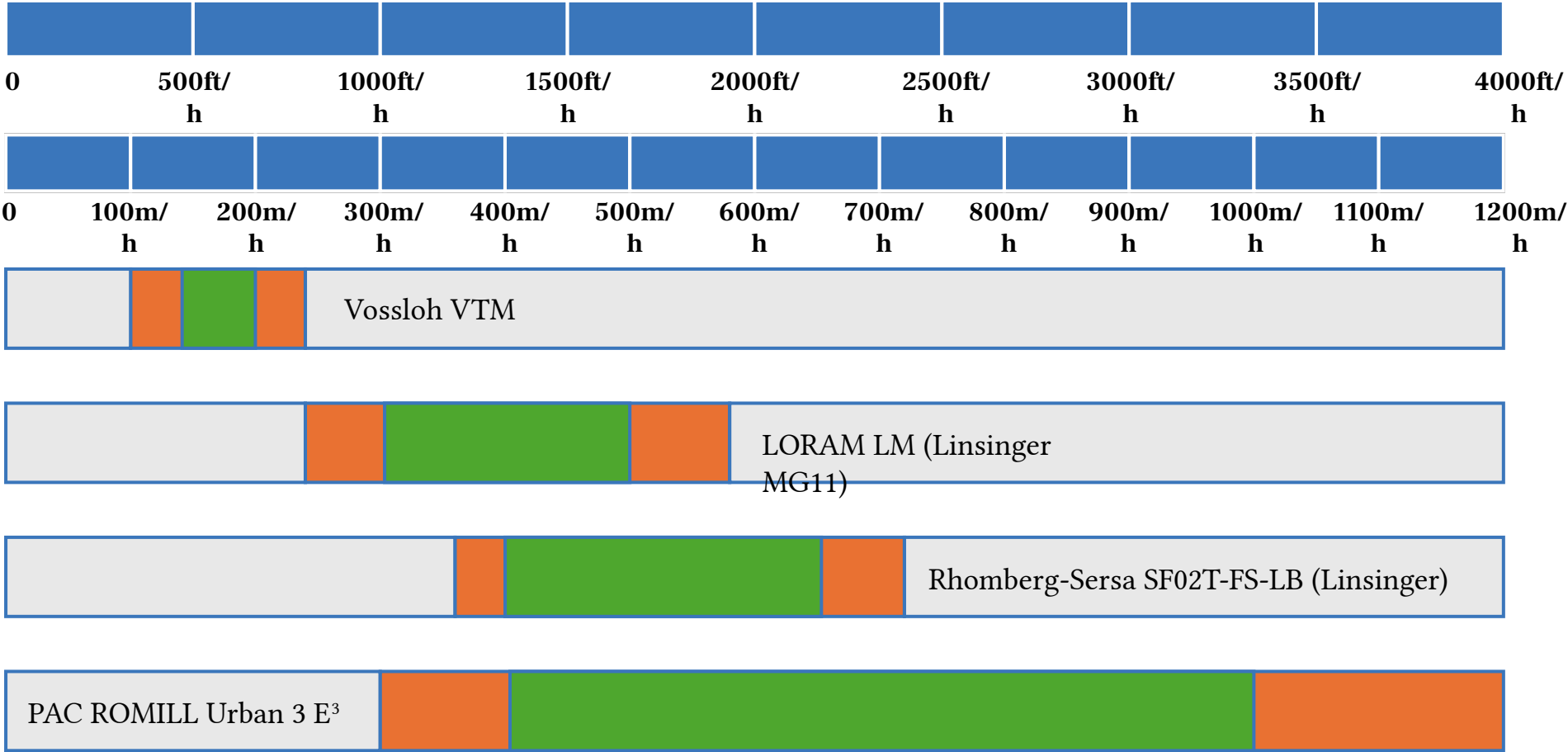
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- Two different speeds
 - Speed per pass (ft/h, mi/h – m/h, km/h)
 - Finished track per hour (ft/h, mi/h – m/h, km/h)
- Speed per pass
 - Transit milling machines: 328ft/h – 3937ft/h (100m/h – 1200m/h)
- Finished track per hour
 - Typically lower than speed per pass



Speed per Pass Overview

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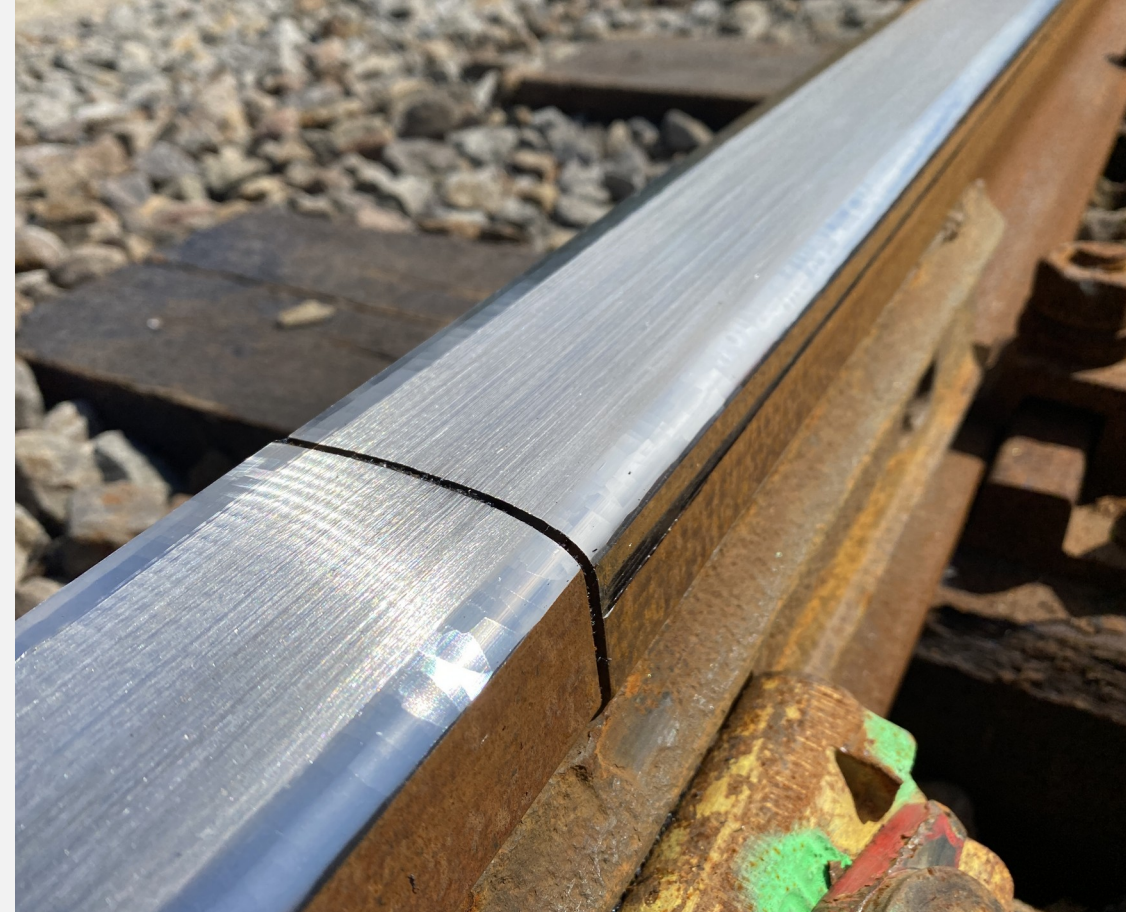




Finished track per hour

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- Processing speed per pass
 - Milling machine and technology
 - Time for ramp-in and ramp-out
- Required metal removal per pass and number of passes
 - Initial profile shape
 - Surface damage condition
- The more metal to remove per pass (area) the slower the pass

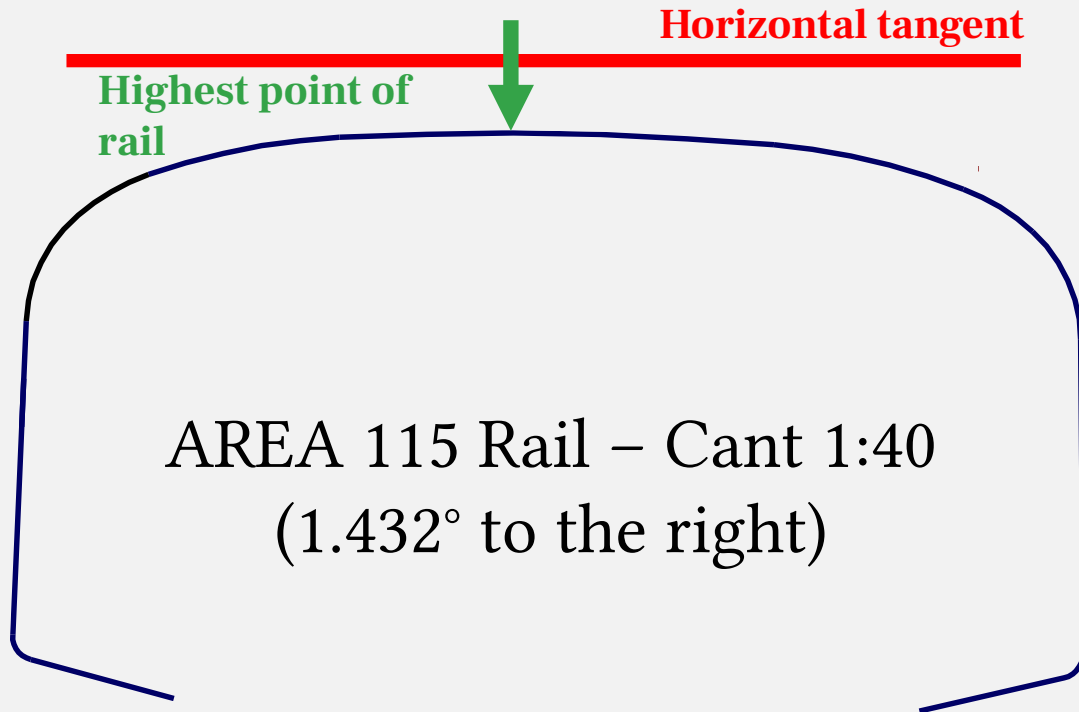




Metal removal (per pass)

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- Material removal in reference to highest point of rail (HPR)

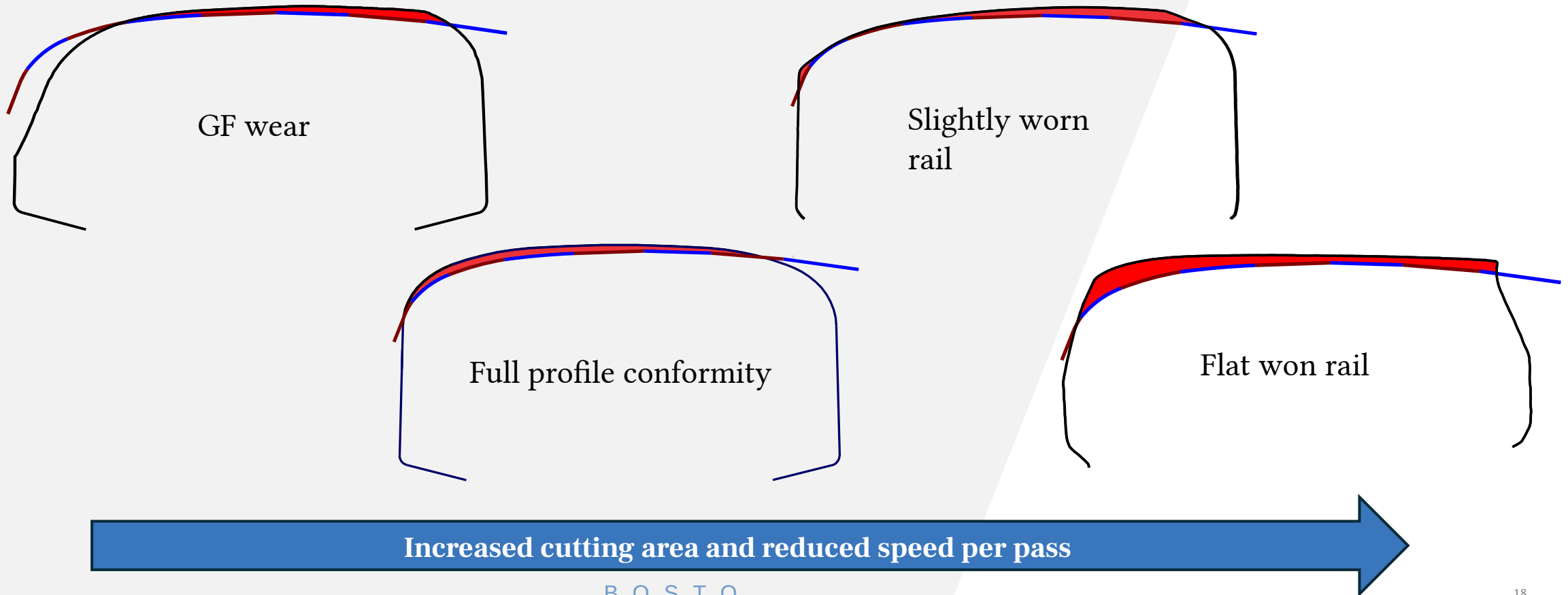




Cutting Depth vs. Area Removed

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- Examples: Same cutting depth – different area (red) removed





Metal Removal and Grinding

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- Grinding stones aligned across transversal profile
 - Profile created as polygon
- 12 stones per rail required to cover full profile in one pass
 - Dependent on acceptable facet width for final pass
- Metal removal per pass
 - Uniform metal removal across full profile vs. localized higher removal

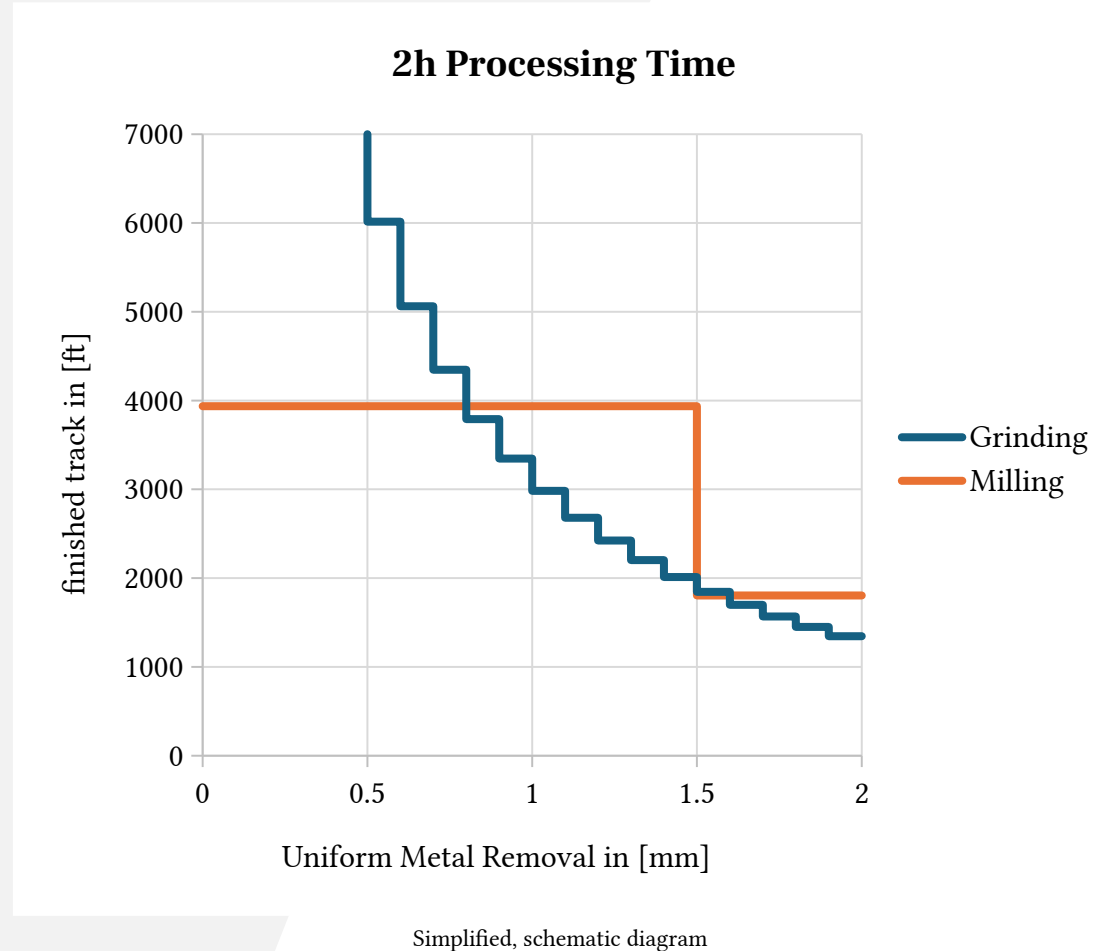




Example: Finished track in 2h

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- 2 h of total processing time window
- Uniform Metal removal
 - Grinding 3.73 mph and 0.1mm
 - 12 stones per rail for full profile
 - Milling 0.373 mph and 1.5mm
- Grinding: 2 min for directional change
- Milling: 10 min to set back
- Cut-off: 9 passes / 0.8mm

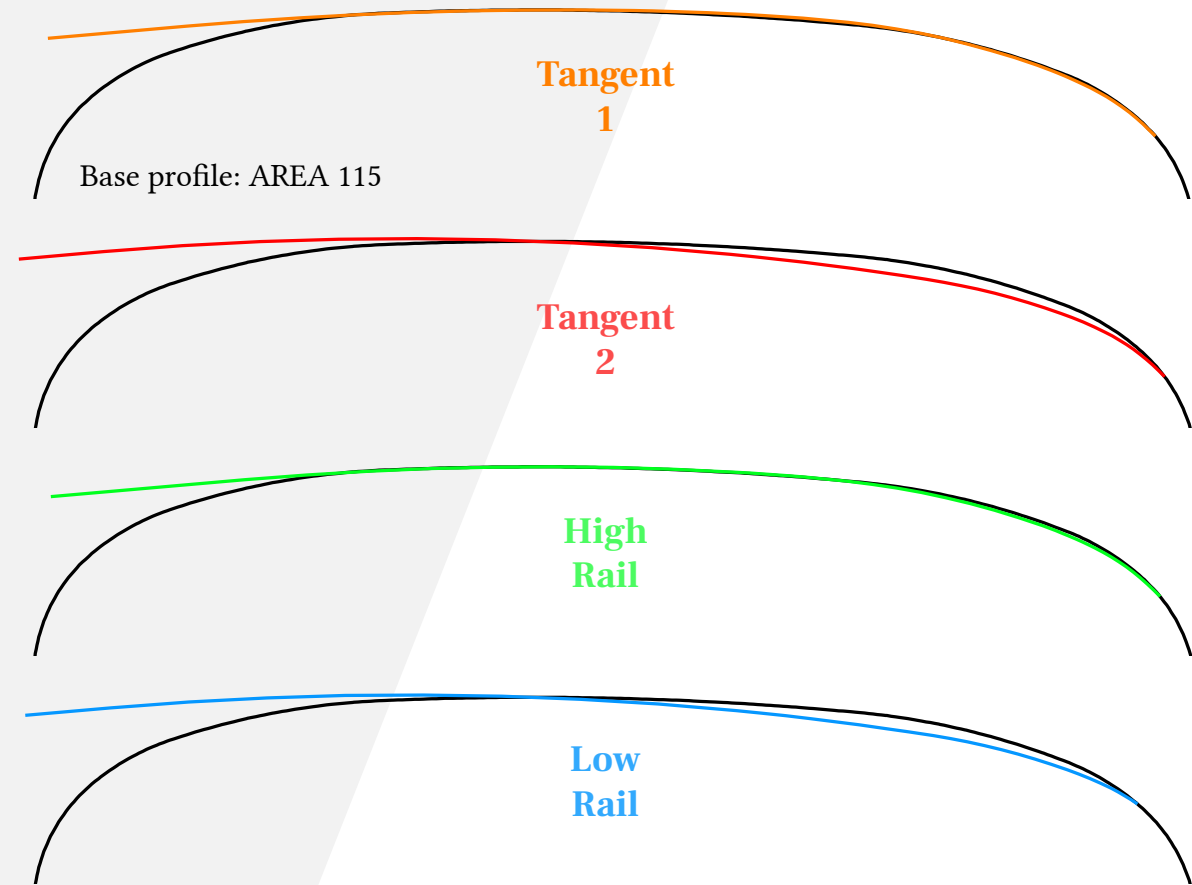




Target Profiles

- Significant difference between Europe (1-2 profiles) and North America (multiple)
- Transits typically have 4 target profiles
 - 2 tangent profiles – distribute wheel wear over a greater area
 - 50% - 50%
 - One high rail and one low rail profile

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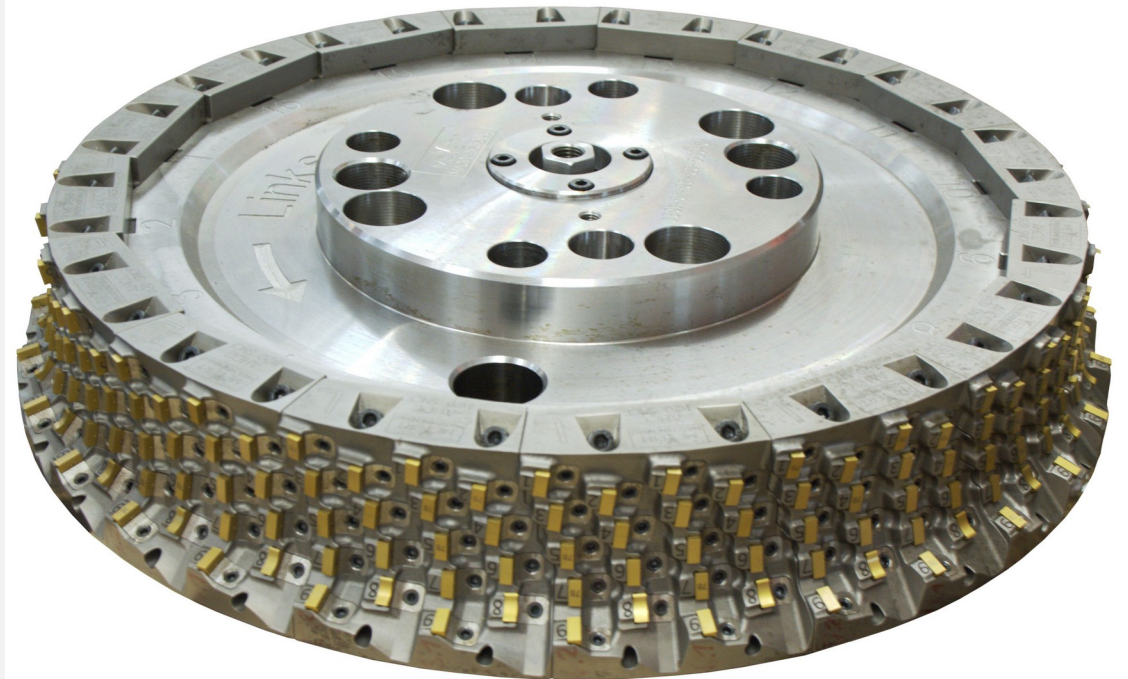
Example Profiles



Target Profiles and Milling

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- Every profile needs an according pair of cutter heads
- Change cutter heads to change to a different profile
 - Cassettes or full cutter head
- Cutter head change: 10-15 min
 - Out of 2h milling time: 8-12%





Target Profiles Creation

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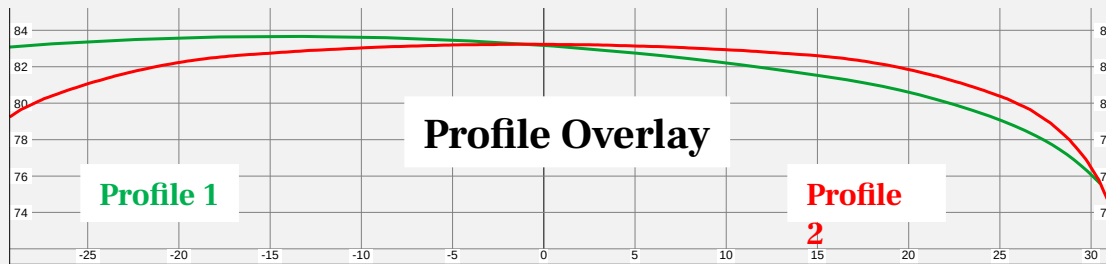
- Target profiles vs. creation accuracy
 - Best case: $\pm 0.3\text{mm}$ for grinding, $\pm 0.2\text{mm}$ for milling
 - Specification?
- Specification vs. quality control
 - Star Gauge or Bar Gauge limited measurement precision
 - Electronic measurement technology



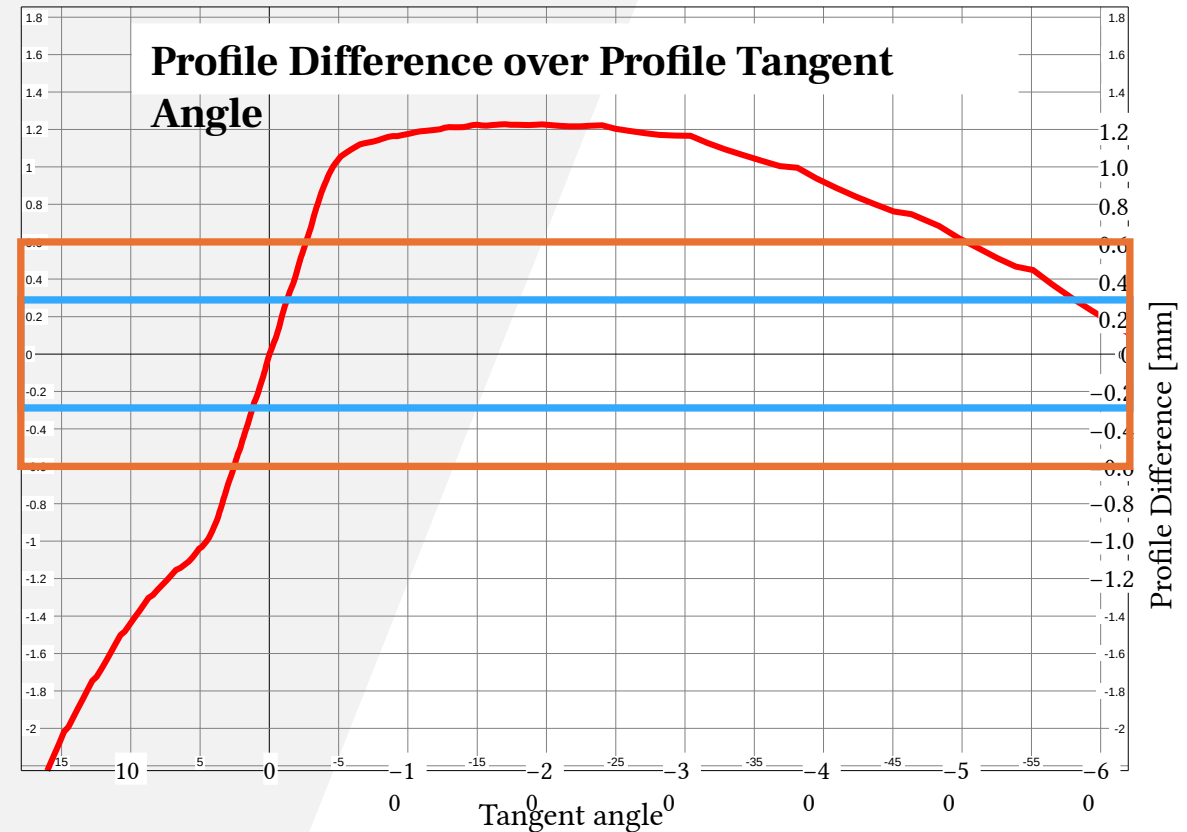


Example: Significant Profile Difference

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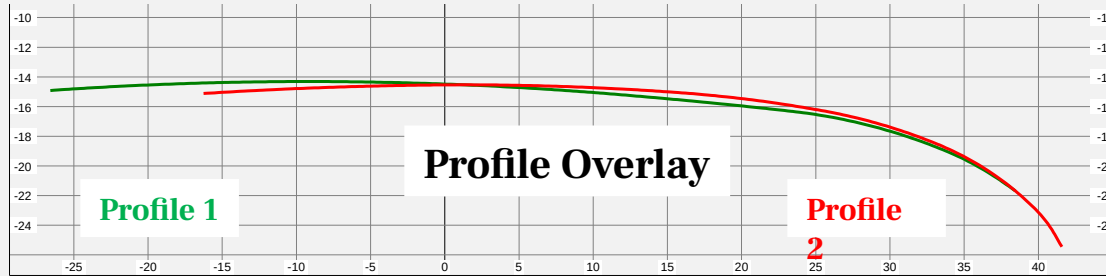


- 2 target profiles with significant difference
- Profiling Specifications
 - +/- 0.3mm ok
 - +/- 0.6mm ok

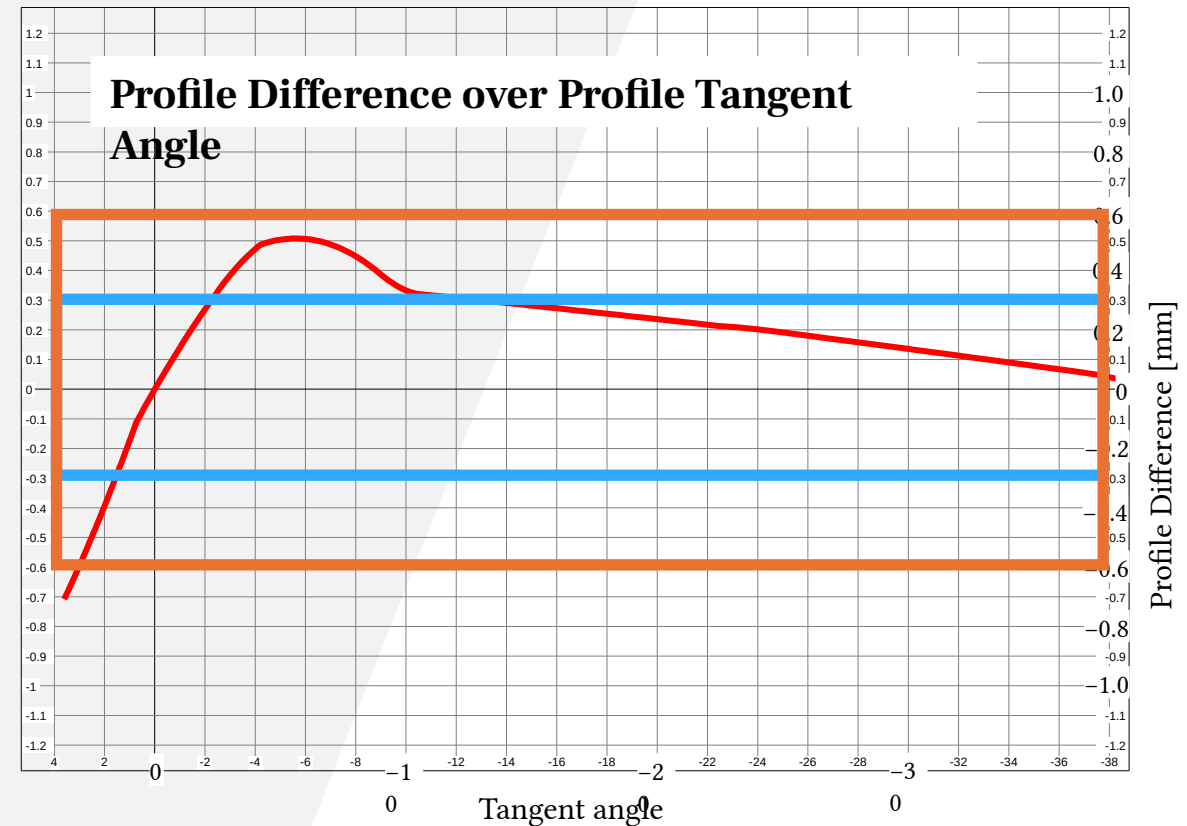


Example: Medium Profile Difference

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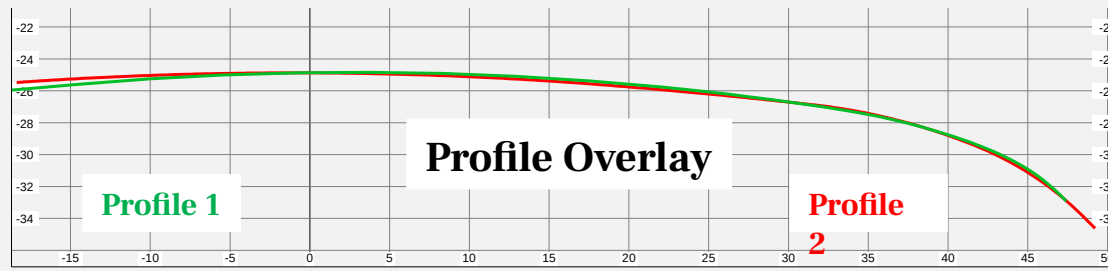
- 2 target profiles with medium profile difference
- Profiling Specifications
 - +/- 0.3mm ok
 - +/- 0.6mm not feasible



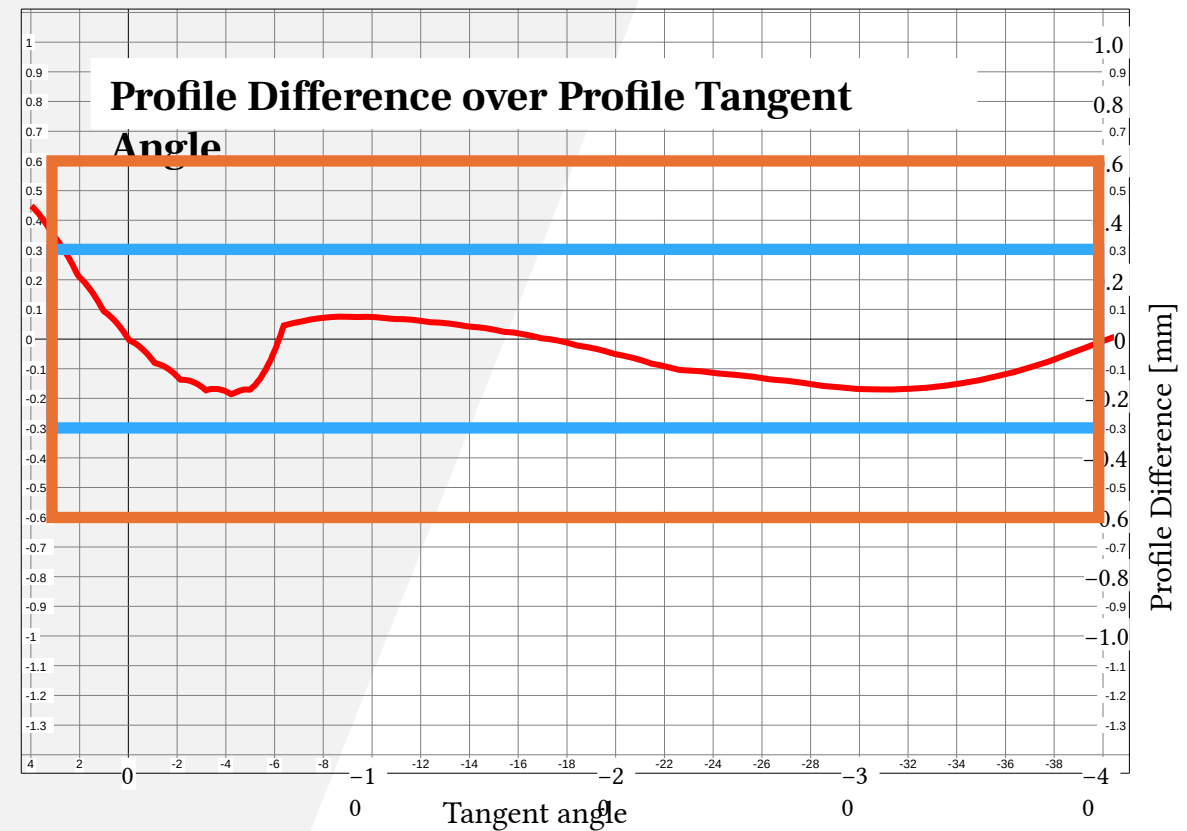


Example: Minor Profile Difference

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- 2 profiles with minor profile difference
- Profiling Specifications
 - +/- 0.3mm not feasible
 - +/- 0.6mm not feasible

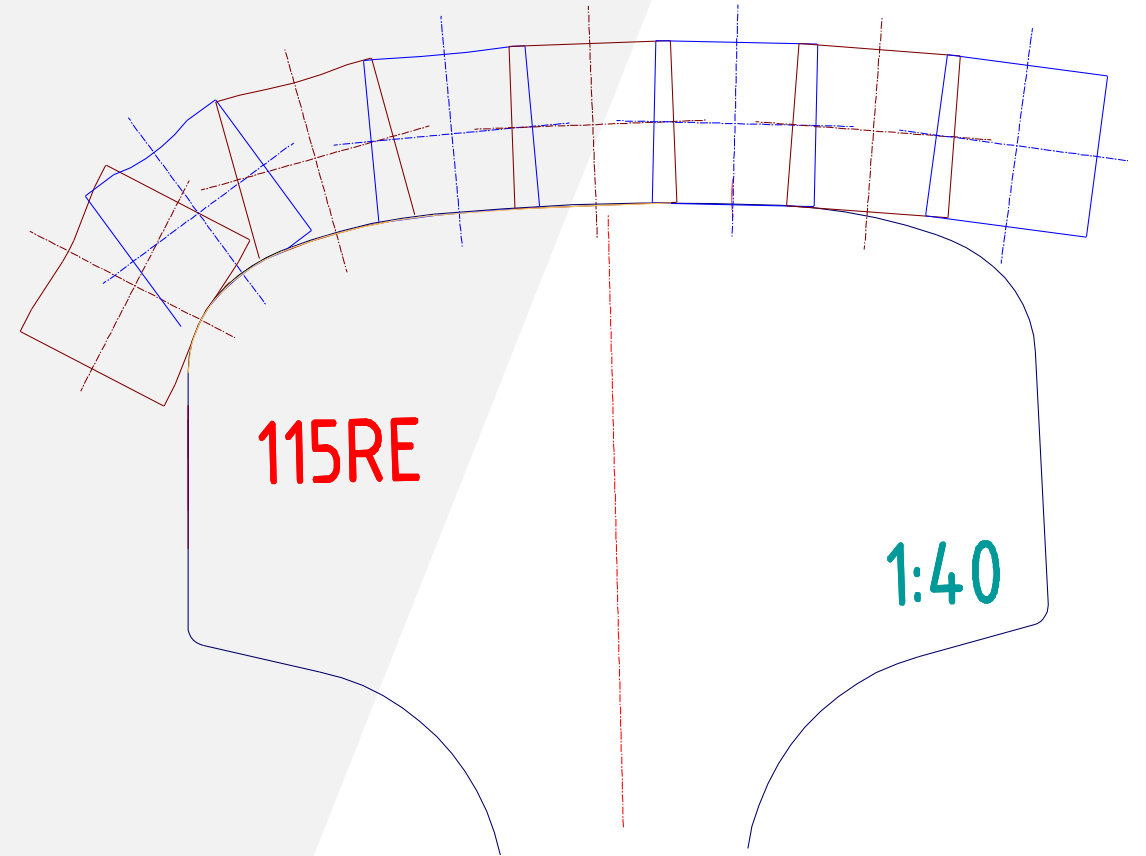




Target Profiles and Milling

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- Focus/strength of rail milling:
 - Removal of (deep) damage
 - Correction of severely deformed profiles
- Milling: 1-2 target profiles
 - Benefit from damage removal higher than getting the profile 100% right
 - Additional benefits
 - Maximized Productivity
- Potentially: touch up with grinding at a later stage





Additional Benefits and their Value

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- Dust free / low dust
 - Only polishing may create limited dust — collection system
 - Air quality in tunnels, stations, next to the machine
- Spark free / low spark
 - Minimal fire risk
 - No damage to nearby objects
- Battery drive
 - Zero Exhaust / NOx / CO
- Challenge: \$\$\$ Value





Technology Specific Requirements

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- Different requirements for milling and grinding
- Grinding:
 - Spark and Dust-handling, stone change, stabling location and access to machine
- Milling:
 - cutter head / cassette change, chip-bunker evacuation, milling insert indexing, work-shop container, stabling location and access to machine





How to Translate This Into RFQs



The Technical Starting Point

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- What is your current track condition
 - If you do not know, get support
 - Independent, knowledgeable, reliable third party that you trust
- What is the required work
 - A shiny rail
 - Preventive work
 - Corrective work
 - Regenerative work





Technical RFQ Compilation

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- Use previous RFQs as starting point
 - Do not copy and paste without review
- Specific technology vs technology open: milling and/or grinding
- If you do not have in-house capability, get support.
 - Independent, knowledgeable, reliable third party that you trust





Technical Work Requirements

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- Clearly define your requirements
 - Type of maintenance
 - How many shifts
 - Target profiles, accuracy, surface quality and quality control
 - Additional requirements
- Provide as much information about your track condition as possible





Tender Review Process

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- Price vs. Budget
- Not just price but what gets done with this money
 - Machine performance vs. price
- Definition of machine performance
 - Finished track/h
 - The clearer your tender definition the easier the comparison





Full Circle: Finished track per h

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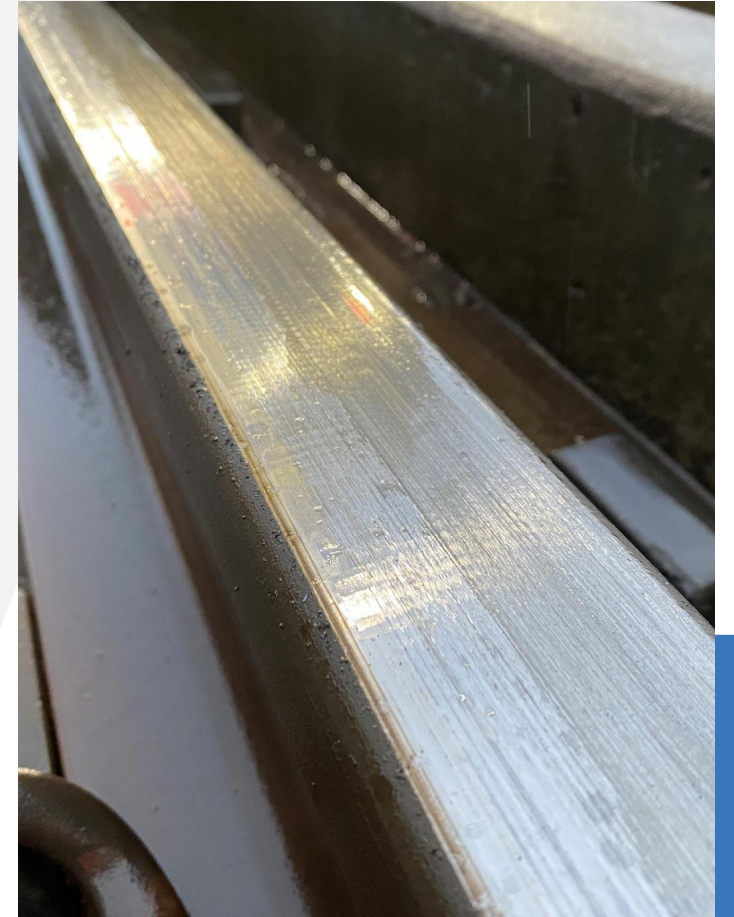
- Influencing Factors
 - Technology
 - Metal Removal + Definition
 - Speed per pass
 - No. of passes
- According to RFQ Specification
 - Transversal and long. profile, profile accuracy, surface quality, damage removal
 - Quality Control and measurement precision
- Comparability





Well Maintained Rail

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Summary

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- A Supplier's View
 - Milling and Grinding
- Importance of finished track per hour
 - Technology specific influencing factors
 - According to Specifications
- Importance of well-defined Scope of Work in RFQ



A close-up, low-angle shot of a train track. A blue and yellow train car is visible in the distance, moving away from the viewer. The tracks are made of steel rails and wooden ties, with gravel ballast. The background is slightly blurred, emphasizing the train car.

Thank You for Your Attention

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